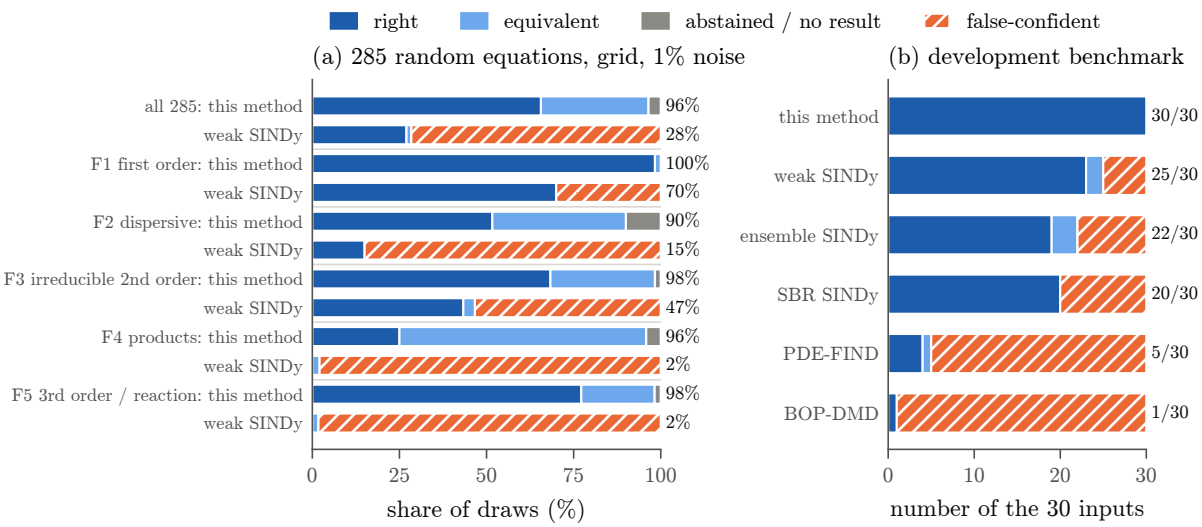
right
equivalent
abstained / no result
false-confident
(a) 285 random equations, grid, 1% noise
all 285: this method
weak SINDy
F1 first order: this method
weak SINDy
F2 dispersive: this method
weak SINDy
F3 irreducible 2nd order: this method
weak SINDy
F4 products: this method
weak SINDy
F5 3rd order / reaction: this method
weak SINDy
96%
28%
100%
70%
90%
15%
98%
47%
96%
2%
98%
2%
0
25
50
75
100
share of draws (%)
(b) development benchmark
this method
weak SINDy
ensemble SINDy
SBR SINDy
PDE-FIND
BOP-DMD
30/30
25/30
22/30
20/30
5/30
1/30
0
10
20
30
number of the 30 inputs